

MATH

Courses

The Mathematics curriculum is traditional, although many of the best practices for mathematics classrooms (technology, group projects, inquiry-based learning, writing mathematics, etc.) are being incorporated where appropriate to enhance the traditional topics.

The following list of courses represents current or recent course offerings. See the [course catalog](#) for updated information.

52-064 Modern Quantitative Methods

This course focuses on modern quantitative and mathematical techniques, and illustrates ways in which these methods are used in the real world. Students will develop number sense, improve their visual/spatial reasoning, and better understand important mathematical functions. An emphasis is placed on visualizing and graphing functions, manipulating mathematical expressions, and interpreting models. Topics studied may include linear, polynomial, exponential, logarithmic, and trigonometric functions. Students who may benefit from additional preparation prior to taking Mathematics 52-164 Modern Calculus I are encouraged to take this course. See math placement guidelines. Cannot be taken for credit after completing Mathematics 52-164 Modern Calculus I with at least a C-, or concurrently with Mathematics 52-164 Modern Calculus I. This course may not be used for the Mathematics or Computational Mathematics majors or minors. (Fall, Spring) (NS)

52-104 Explorations in Mathematics

This course presents the spirit and beauty of mathematics through topics chosen by the instructor, emphasizing the role that mathematics plays in society. Topics may include mathematics in art and literature, Euclid's Elements, game theory and voting theory. The mathematical content may include geometry, algebra, and number systems. The course is suitable for a general audience with a broad spectrum of backgrounds and abilities and also satisfies requirements for EC-6 or 4-8 teacher certification. This course may not be used for the Mathematics major or minor. (NS)

52-114 Introduction to Statistics

This course provides students in the social and natural sciences with the skills necessary to perform elementary statistical analysis. Topics include descriptive measures, sampling theory, Student-T and normal distributions, estimation and hypothesis testing with p-values, regression and correlation. This course may not be used for the Mathematics major or minor. Contributes to Data Analytics, Data Science, and Health Studies. (Fall, Spring) (NS)

52-164 Modern Calculus I

This course focuses on introducing calculus with a modeling first approach. Topics include: functions as models of data, vectors, differential calculus of functions of one and several variables, optimization, and integration. Applications may be drawn from varied areas, such as biology, chemistry, economics, and

benefit from additional preparation prior to taking this course are encouraged to take Mathematics 52-064 Modern Quantitative Methods. Before registration, consult the math placement guidelines on the registrar's website. (Fall, Spring) (NS)

52-204 Topics in Mathematics

This course investigates a topic in Mathematics that varies according to the interests of the professor. This course may be repeated with a change in the topic. (NS)

52-264 Modern Calculus II

This course focuses on calculus useful for both theoretical and applied work in the mathematical, natural, and social sciences. Topics include: introductory differential equations, sequences and series, Taylor series, techniques and applications of integration, partial derivatives, multiple integration, and limits. Attention is given to both symbolic and numerical computing. Prerequisite: Mathematics 52-164. (Fall, Spring) (NS)

52-291 Putnam Power Hour

This course is designed to sharpen problem solving abilities. Students will tackle challenging problems from the William Lowell Putnam Competitions of previous years and study some of the published solutions. Students enrolled in this course will be encouraged to compete in the Putnam Competition in early December. This course may be repeated for credit, but may not be counted toward the major or minor, and must be taken Pass/D/F. Prerequisite: Consent of instructor.

52-304 Selected Topics

May be repeated with change in topic. Prerequisite: Permission of instructor.

52-364 Modern Calculus III

This course focuses on calculus useful for the mathematical and physical sciences. Topics include: scalar and vector-valued functions and derivatives; gradients, contour plots, and constrained and unconstrained optimization of multiple variables; integration over regions in various coordinate systems; parameterization and integration over curves, and surfaces; divergence; curl; Green's theorem; and formal mathematical definitions of the integral, derivative, and limit. Prerequisite: Mathematics 52-264. (Fall, Spring) (NS)

52-384 Discrete Mathematics

See Computer Science 54-384. (Fall) (NS)

52-404 Geometry

This course investigates various approaches to geometry. Topics may include synthetic geometry, analytic geometry, projective geometry, differential geometry, Euclidean geometry and non-Euclidean geometry. Prerequisite: Permission of instructor. (Fall, odd years) (NS)

52-414 Operations Research

See Computer Science 54-414 and Business 30-414. Contributes to Data Analytics and Data Science.

52-524 Introduction to Numerical Analysis

This course investigates the derivations and applications of numerical techniques. Topics include: interpolation, approximation, numerical differentiation and integration, zeros of functions and solution of linear systems. Also Computer Science 54-524. Prerequisites: Mathematics 52-264, 52-674, and

This course is a calculus-based, mathematical introduction to the fundamental principles of probability theory and applications. Topics include combinatorial analysis used in computing probabilities, the axioms and properties of probability, conditional probability, independence of events, discrete and continuous random variables, the standard distributions, expected value and variance, joint distributions, distributions of a function of a random variable, and sampling distributions. Also included are theoretical results such as Bayes' Theorem, Central Limit Theorem, Law of Large Numbers, the Empirical Rule, Hypothesis Testing and Confidence intervals at least for a single mean and a single proportion. Contributes to Data Analytics and Data Science. Prerequisite: Mathematics 52-264. (Spring) (NS)

52-674 Linear Algebra

This course is an introduction to the basic structure of proofs, linear equations and matrices, vector spaces, linear mappings, determinants, eigensystems, orthogonality, matrix decompositions and applications of linear algebra. Prerequisite: Mathematics 52-264, or permission of instructor. (Fall) (NS)

52-684 Algebraic Structures

This course investigates the theory of sets, relations, functions, groups and rings. A rigorous approach to learning and writing proofs is emphasized. Prerequisite: Mathematics 52-674 or permission of instructor. (Fall) (NS)

52-754 Differential Equations

This course investigates the theory and application of differential equations. Topics include both linear and nonlinear first order ordinary differential equations, numerical solutions, and higher order linear ordinary differential equations. Solution techniques may include undetermined coefficients, variation of parameters, power series solutions, and Laplace transforms. Additional topics may be chosen from linear systems, nonlinear systems and Fourier series analysis of partial differential equations with boundary conditions. Prerequisite: Mathematics 52-364, or permission of instructor. (Spring) (NS)

52-834 Complex Analysis

This course investigates functions of a complex variable. Topics include algebra and geometry of complex numbers, analytic and harmonic functions, mappings, Taylor and Laurent series, and contour integration. Prerequisite: Mathematics 52-364, or permission of instructor. (Fall, even years) (NS)

52-844 Seminar in Special Topics

This course is a limited enrollment seminar in a major area of mathematics not generally covered in other courses. Topics may include but are not limited to advanced analysis, combinatorics, logic and history of mathematics. The course may be repeated for credit as topics vary. (NS)

52-854 Real Analysis

This course investigates the algebra and topology of the real numbers. Topics include completeness, sequences, limits and continuity, differentiation, the Mean-Value Theorem, Taylors Theorem and infinite series. May also include sequences and series of functions. A rigorous approach to learning and writing proofs is emphasized. Prerequisite: Mathematics 52-674, or permission of instructor. (Spring) (NS)

52-894 Senior Seminar in Mathematics

This course will fulfill the capstone requirement in Mathematics. Since it serves as a culmination of the student's undergraduate mathematical experience, a balance is sought between application and theory.

A major semester project resulting in a written paper and an oral presentation is required, an external presentation may also be required. Prerequisites: Five courses in the major at the 300 level or above, Computer Science 54-184, and permission of instructor. (Fall) (NS) (WA)

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CONTACT